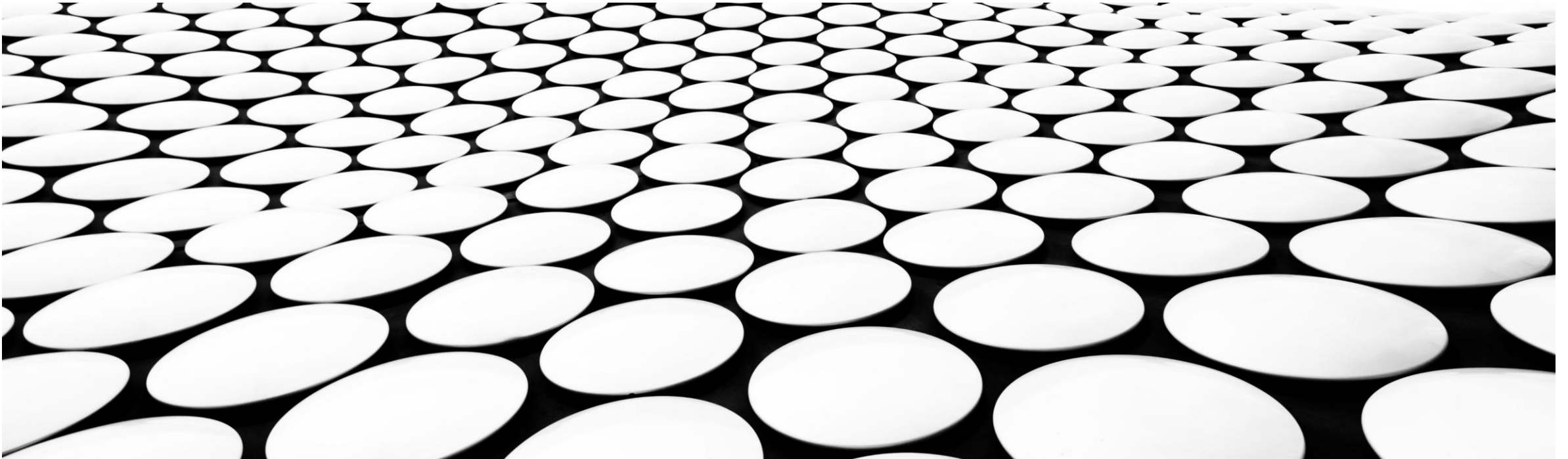


# CSI4142 – FUNDAMENTALS OF DATA SCIENCE

COURSE ORGANIZATION, LEARNING OBJECTIVES AND COURSE REQUIREMENTS





## GOALS

- Course basic information
- Course calendar description
- Course objectives
- Course topics / schedule
- Course evaluation
- Course rules

**Professor**

Caroline Barrière

**Email**

[cbarrier@uottawa.ca](mailto:cbarrier@uottawa.ca)

**Term**

Winter 2026

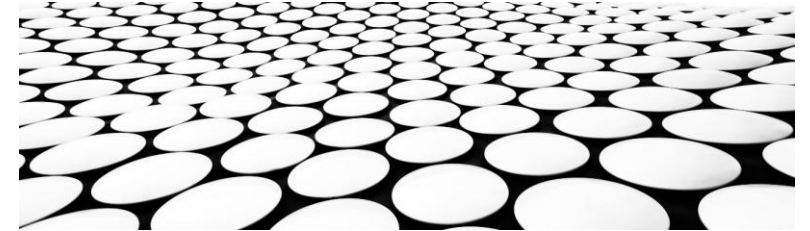
**Course hours / location**

Wednesday, 10:00-11:20, in CBY C03

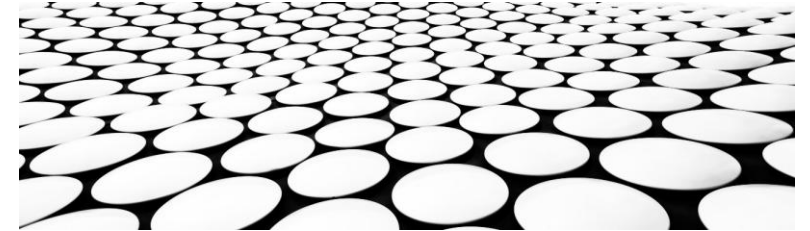
Friday, 8:30-9:50, in CBY C03

**Communication**

- I will use the announcement feature in Brightspace to communicate with you.
- Please use the Brightspace Forum to ask questions, so that everyone else in the class can benefit from my answers.
- Please communicate with me by email only for personal issues.

**COURSE INFORMATION**

## COURSE INFORMATION

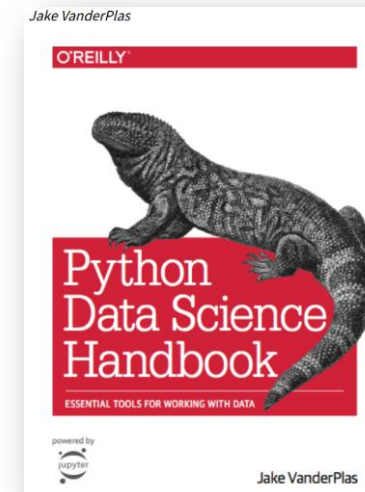


### Course material

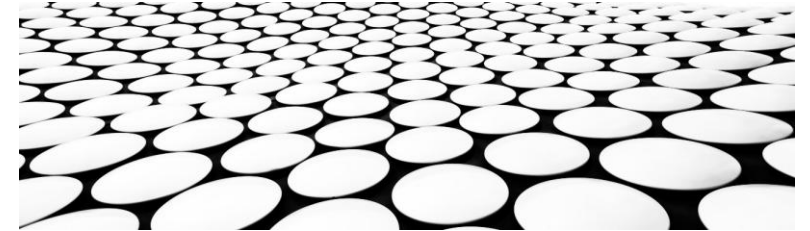
- There is no mandatory textbook for this course.
- I will provide:
  - PDF from the slides presented in class
  - A list of additional material, in the form of articles or online videos.

### Python Programming resources

- You can find many online resources to help you with Python coding
- Here is one suggestion:
  - <https://jakevdp.github.io/PythonDataScienceHandbook/>
  - <https://github.com/jakevdp/PythonDataScienceHandbook?tab=readme-ov-file>



## COURSE INFORMATION



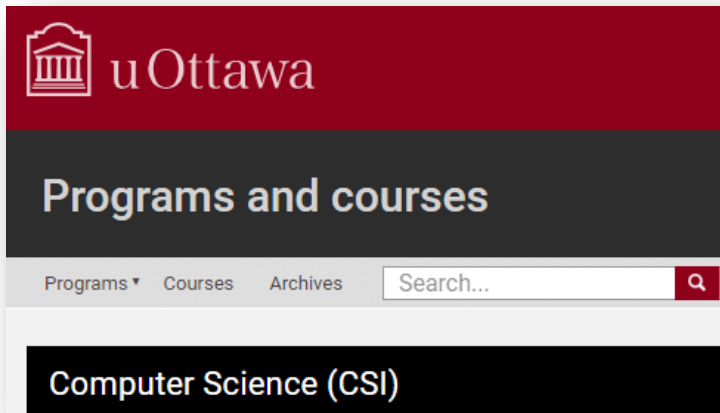
### Course attendance

- This is an *in-person* class, so attendance is highly recommended
- Classes are not recorded, but PDF of slides are provided on Brightspace

### Consultation hours

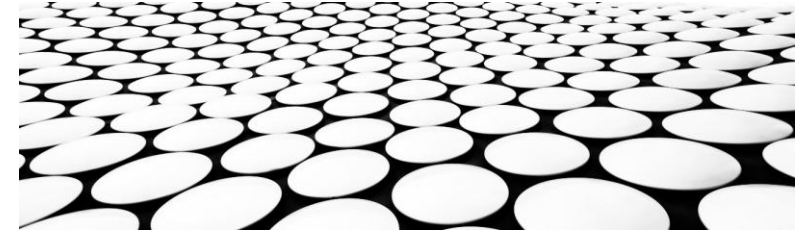
- I will regularly monitor the discussion forum in Brightspace.
- At the end of every class, I can be available 15 min. for questions.
- All programming questions should be asked to the TAs on Brightspace.

I will not answer  
coding questions



We will cover these topics,  
but not all with the same  
depth

## CALENDAR DESCRIPTION



### CSI 4142 Fundamentals of Data Science (3 units)

Big data, analytics, and cloud computing; data preparation: organization, basic statistics, cleaning, and integration; data mining techniques: pattern mining, classification, clustering, outlier and anomaly detection; model evaluation; data warehousing and multi-dimensional analysis; data visualization and visual data analytics.

**Course Component:** Lecture

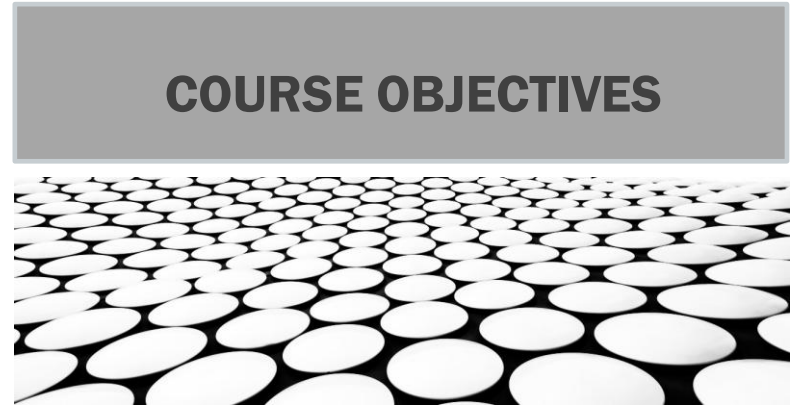
**Prerequisites:** [CSI 2132](#), ([CSI 3120](#) or [SEG 2106](#)), [MAT 2377](#) or ([MAT 2371](#) and [MAT 2375](#)).

### Gain expertise in the area of Data Science

Data Science is a large and complex interdisciplinary field which is impossible to cover fully within a 12 weeks semester. The main focus will be on Data Analytics. Other topics will be briefly touched upon.

*I hope for each student, at the end of this semester to:*

- Understand the value of Data Science and Data-Driven Decision Making in Industry
- Be able to perform Exploratory Data Analysis anchored in mathematical foundations and showed using various visualization approaches
- Understand the various issues around data collection and usage (biases, errors, etc) as well as the importance of data cleaning
- Be able to perform Predictive analysis using supervised machine learning
- Be able to perform data segmentation and recommendations based on unsupervised machine learning approaches
- Be able to perform time series analysis and understand its importance in data science
- Understand the value of data engineering and data warehousing for future work in collaboration with data engineers
- Understand challenges for big data analysis



Assignments will cover these topics.

## COURSE OBJECTIVES

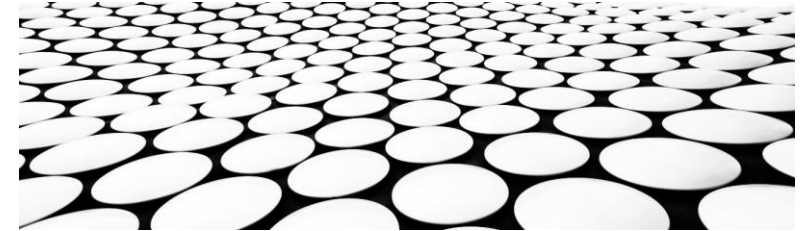
### Develop your skills for using Generative AI for enhancing your learning

We will sometimes use Generative AI **during lectures** for:

- Finding focused examples (help us explore different domains)
- Understanding the difference between concepts (help us find criteria for comparison)

You can use Generative AI for **your assignments**:

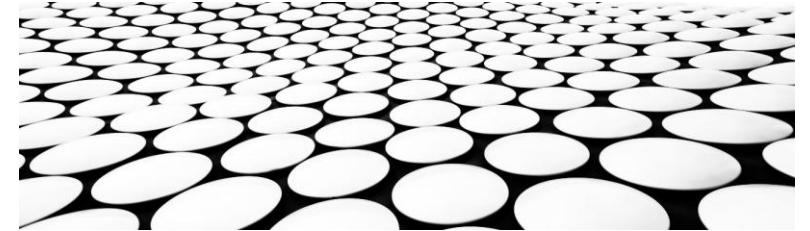
- Obtaining starter code that you can modify
- Obtaining help on specific methods and their parameters that you can adapt to your context





## Tentative schedule with topics covered

### COURSE TOPICS / SCHEDULE

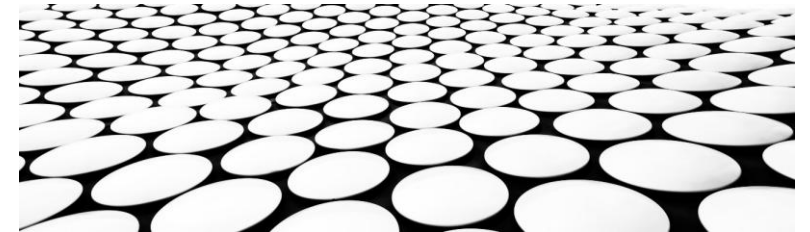


| Week | Dates (W/F)           | Topic                                                                                                      |
|------|-----------------------|------------------------------------------------------------------------------------------------------------|
| 1    | <i>January 14/16</i>  | Course organization / Overview of covered topics<br>Introduction to Data Science / Data types and Insights |
| 2    | <i>January 21/23</i>  | Data Science Process, Descriptive statistics, Exploratory Data Analysis (EDA), Visualization               |
| 3    | <i>January 28/30</i>  | Data collection and data sources, Inferential statistics                                                   |
| 4    | <i>February 4/6</i>   | Data cleaning (duplicates, validation, outliers)                                                           |
| 5    | <i>February 11/13</i> | Review Session – Quiz 1                                                                                    |
|      | <i>February 18/20</i> | READING WEEK                                                                                               |

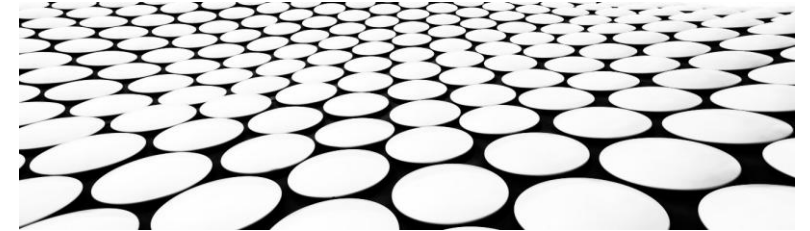
## Tentative schedule with topics covered

| Week | Dates (W/F)                | Topic                                                                                                                          |
|------|----------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| 6    | February 25/27             | Missing data (Imputation) and biases                                                                                           |
| 7    | March 4/6                  | Predictive analysis, Model building and Evaluation<br>Supervised learning, classification (linear regression + decision trees) |
| 8    | March 11/13                | Predictive analysis (continue): ensemble models + feature engineering                                                          |
| 9    | March 18/20                | Unsupervised learning, clustering, recommendation systems                                                                      |
| 10   | March 25/27                | Review Session – Quiz 2                                                                                                        |
| 11   | April 1<br>(Easter Friday) | Time Series (Forecasting)                                                                                                      |
| 12   | April 8/10                 | Time series (Various analysis), Integration (Data warehousing, ETL)                                                            |
| 13   | April 14 (make up)         | Integration (Visualization)                                                                                                    |

## COURSE TOPICS / SCHEDULE



## COURSE EVALUATION



### Evaluation in 3 components:

- Programming assignments (40%)
- Quizzes (or Take-Home) (16%)
  - Quiz 1 (8%) – Friday, February 13, 8:30 am
  - Quiz 2 (8%) – Friday, March 25, 8:30 am
- Final Exam (44%)

### ***PLEASE NOTE***

- The weight of any assignment or quiz CANNOT be transferred to the final exam
- Only justified medical reasons can lead to transfer

## Programming assignments (40%)

### *Goal*

The purpose is to get hands-on experience and gain knowledge of Python tools and packages for data analysis in relation to concepts presented in class.

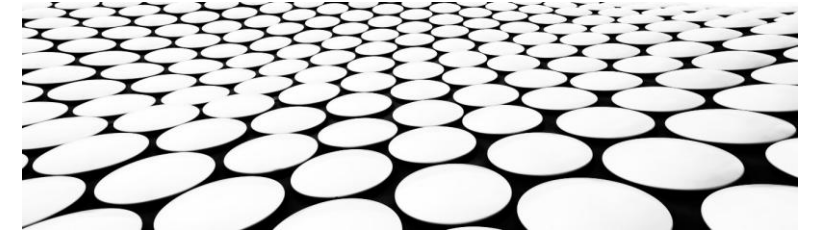
### *Requirements*

- Work must be done in groups of 2
- All assignments need to be programmed in Python
- For each assignment, make sure you read the requirement document carefully.

### *Deliverables*

- All assignments will be submitted as Jupyter Notebooks
- Importance will be put, not only on code, but on clear presentation and explanation of the code and results submitted
- Meetings with TAs will be organized to verbally present/explain your work

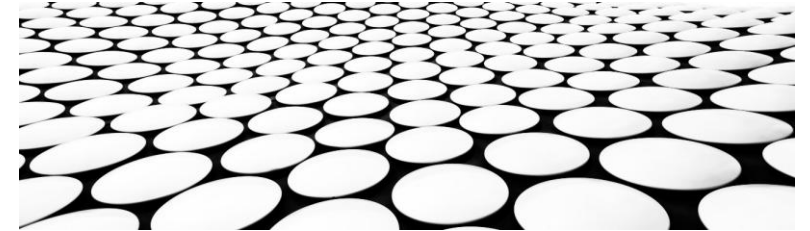
## ASSIGNMENTS



TAs will be available to help you with assignments. They will perform forum monitoring daily.

## Programming assignments (40%)

## ASSIGNMENTS



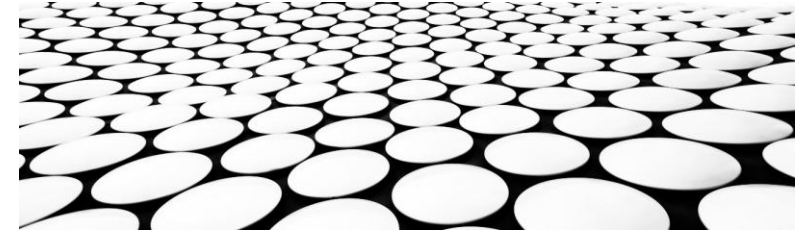
| Assignment   | Value | Due Date   | Topic                                             |
|--------------|-------|------------|---------------------------------------------------|
| Assignment 1 | 10%   | February 3 | Exploratory data analysis (EDA) and Visualization |
| Assignment 2 | 10%   | March 3    | Data wrangling                                    |
| Assignment 3 | 10%   | March 17   | Predictive analysis (supervised learning)         |
| Assignment 4 | 10%   | April 7    | Recommendation (unsupervised learning)            |

*Tentative topics...*

### Quizzes (or take-home) (16%)

- 80-minute exams
- Performed during Friday morning periods (February 13, March 27)
- Format to be determined (depending on room availability)

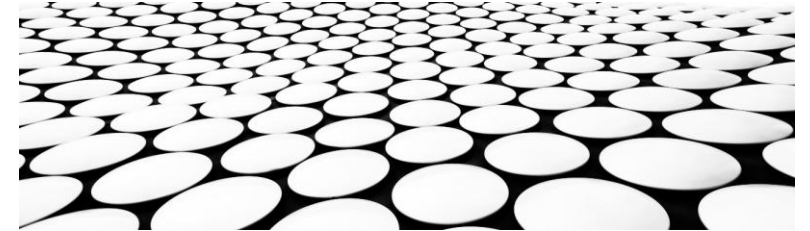
## QUIZZES



### Final exam (44%)

- Performed during exam period (April 17th to 30th)
- Open book
- In Person, on Brightspace
- Plan to bring your laptop

## FINAL EXAM



Open book means access to Brightspace slides and your notes. No other internet access.

## COURSE RULES

### Plagiarism / Honors system

- Please take pride in performing all the work yourself. If you don't know what to do, ask clarification questions on the forum.
- All submissions in YOUR Brightspace should be your own work.
- Any student work largely resembling another student work will be given zero, and students will receive a warning. If there is a second occurrence of copied work, students will receive zero for the semester.

### Late work

- Any late assignment will be penalized 20% per day.

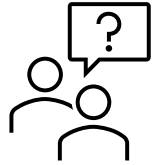
### Course attendance request

- Given my hypersensitivity to noise, I please ask everyone attending to not hesitate to raise their hand and ask me questions, but please DO NOT discuss with your neighbour as I CANNOT continue teaching when students talk to each other.

### Minimum requirement rule

- *A minimum of 50% of the Final Exam is required to pass this course.*





## QUESTIONS?

- Post your questions on the Discussion forum, within the Course Organization + Lectures topic. This is the best option since my answer reaches other students.
- For personal questions, you can also send me an email at [cbarrier@uottawa.ca](mailto:cbarrier@uottawa.ca).